

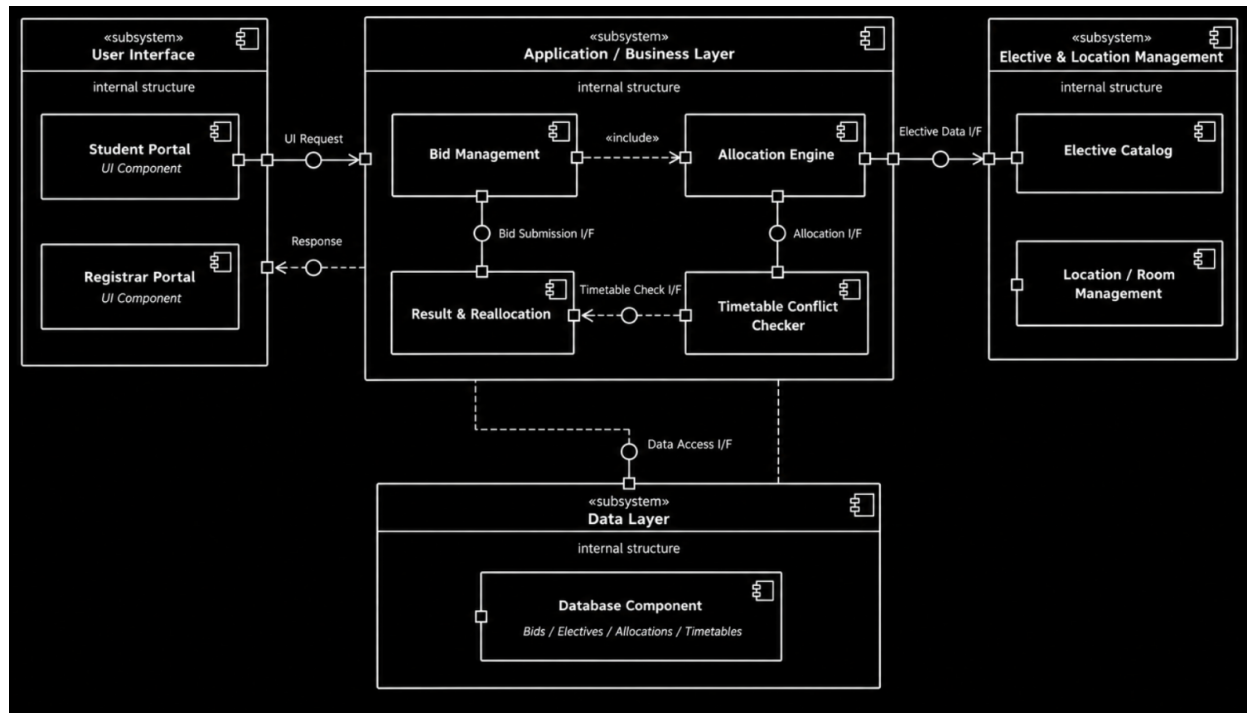
Software Engineering

Lab 3 – Architecture Selection & Component Diagram Write-up

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1. Component Diagram



2. Justification

Architecture Selection: I chose Layered (N-Tier) Architecture for the Academic Elective Bidding & Allocation System.

Architectural Choice

- I split my system into clear layers/subsystems — User Interface, Application/Business Layer, a supporting Elective & Location Management module, and the Data Layer. Each layer only talks to the one next to it, which is the basic idea of a Layered (N-Tier) Architecture.

Two Reasons (Scenario-Related)

- I designed it to match the natural flow of my process: a student bids → my Bid Management module checks it → my Allocation Engine and Timetable Conflict Checker decide the outcome → the result is saved. Since my system already works step-by-step, splitting it into layers keeps my code organized the same way it naturally works.
- I have two different users sharing one logic: both my Student Portal and Registrar Portal send requests through the same UI layer into the same Business layer. This lets both user types reuse the same bidding/allocation rules and the same database, so I don't have to write separate logic for each portal.

Security Advantage

- In my design, students and registrars never touch the database directly. I made sure all requests must pass through my Application/Business Layer first, and only that layer is allowed to talk to my Data Layer through the "Data Access I/F". This means a student cannot directly edit their bid, allocation result, or another student's data — every change is checked and validated by my business rules before it reaches the database.

Performance Benefit

- Because I gave each layer one job, my Allocation Engine and Timetable Conflict Checker can process bids and check clashes in the Business Layer itself, without repeatedly calling the database for every small check. I only access my Data Layer when a final read/write is needed, which reduces unnecessary database load — this matters a lot during peak bidding time when many students submit bids together.